

The relationships between aerobic fitness, power maintenance and oxygen consumption during intense intermittent exercise

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This study examined the relationships between VO_2max , power maintenance and oxygen consumption during intense intermittent work. Female recreational soccer players were assigned to either a low aerobic power group (LOW, $n=6$, mean (SD) $\text{VO}_2\text{max} = 34.4$ (2.4) $\text{mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$) or to a moderate aerobic power group (MOD, $n=7$, $\text{VO}_2\text{max} = 47.6$ (3.8) $\text{mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$). VO_2 was measured while subjects performed 10 6-s all-out sprints (30-s passive recovery) on a Monark cycle ergometer. LOW and MOD subjects generated similar peak 6-s power ($p=.58$) but MOD had a smaller decrement in power (% DO) over the 10 sprints (LOW vs MOD: 18.0 (7.6) vs 8.8 (3.7) % DO, $p=.02$). The MOD group also consumed significantly more oxygen than LOW in 9 of the 10 sprint-recovery cycles ($p<.05$). Significant relationships were seen between VO_2max and the aerobic response to the sprint-recovery series ($r=.78$, $p=.002$) as well as between VO_2max and % DO ($r=-.65$, $p=.02$), while a non-significant relationship was seen between the oxygen consumed during the sprint-recovery cycles and % DO ($r=-.41$, $p=.16$). Thus, VO_2max appears to be related to both an increased aerobic contribution to sprint-recovery bouts and the enhanced ability of the MOD group to resist fatigue during intense intermittent exercise.

Introduction

Though generally associated with exercise of long duration and low intensity, the aerobic metabolic system is stimulated even in high intensity exercise lasting a few seconds (Gaitanos, Nevill, Brooks & Williams, 1991). For example, the majority of energy required for a single bout of brief (<10 seconds) dynamic maximal exercise is provided through phosphocreatine (PCr) breakdown and glycogenolysis leading to lactate formation (Jacobs, Tesch, Bar-Or, Karlsson & Dotan, 1983). However aerobic metabolism may supply as much as 20% of the energy for the first 10*s of a second sprint (Bogdanis, Nevill, Lakomy, Graham & Louis, 1996b).

When performing repeated all-out 6-s sprints, oxygen consumption increases rapidly at the onset of sprinting (Chamari, Ahmaidi, Fabre, Ramonatxot & Préfaut, 1995) and increases with subsequent sprints (Chamari et al., 1995; Gaitanos et al., 1991; Hamilton, Nevill, Brooks & Williams, 1991), attaining levels that may exceed 70% VO_2max (Hamilton et al., 1991). During intense intermittent exercise, the oxygen consumed between sprints has been

associated with enhanced PCr restoration (Bogdanis, Nevill, Boobis & Lakomy, 1996a), which should result in superior power maintenance on subsequent sprints. As well, any increase in VO_2 during the sprints should increase the energy available for muscle contraction by supplementing energy provided anaerobically. The increased energy should translate into more work during the sprints. In support of this, Hamilton et al. (1991) found a relationship between the aerobic response during repeated sprint-recovery intervals and fatigue in peak power ($r = -.60$).

Adaptations due to aerobic training, such as increased number and size of mitochondria, enhanced capillary density and increased aerobic enzyme activity (Anderson & Hendriksson, 1977; Holloszy & Coyle, 1984) and a faster adjustment of oxygen uptake at the onset of exercise (Hickson, Bomze & Holloszy, 1978) may result in increased oxygen consumption during both the sprint and recovery intervals for aerobically fit athletes, contributing to enhanced power maintenance during repeated bouts. Many coaches and athletes consider aerobic training an important component of their preparation for team sports in the belief that it will enhance recovery from intense intermittent exercise. While some studies have revealed an association between aerobic fitness and enhanced recovery from high intensity exercise (Bogdanis et al, 1996a; Dawson, Fitzsimmons & Ward, 1993; Gaiga & Docherty, 1995; Hakkinen & Myllyla, 1990; Hamilton et al., 1991; McMahon & Wenger, 1998; Yoshida & Watari, 1993), others have failed to confirm the relationship (Bell, Snyder, Davies & Quinney, 1997; Cooke, Petersen & Quinney, 1997) and only Hamilton et al. (1991) considered differences in the aerobic response to the exercise. Unfortunately the comparisons made by Hamilton and colleagues (1991) were between games players and endurance trained athletes who happened to differ in $\text{VO}_{2\text{max}}$, making comparisons on the basis of $\text{VO}_{2\text{max}}$ less precise. It was therefore the intent of this study to examine the relationships between aerobic fitness ($\text{VO}_{2\text{max}}$), power maintenance and oxygen consumption during high intensity intermittent exercise performed by female soccer players differing in $\text{VO}_{2\text{max}}$.

Methods

Subjects

All study procedures were approved by the Human Research Ethics Committee of the University of Victoria. Nineteen female recreational soccer players gave their informed consent prior to participation in the study. Subjects with $\text{VO}_{2\text{max}}$ scores $>43 \text{ mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ were assigned to a moderate aerobic power (MOD) group whereas subjects with $\text{VO}_{2\text{max}}$ scores $<38 \text{ mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ were assigned to the low aerobic power group (LOW). Physical and performance characteristics are presented in Table 1.

Design

Subjects participated in two laboratory sessions over a 2 week period. On day 1, the subjects performed a $\text{VO}_{2\text{max}}$ test followed by familiarisation to the sprint test protocol. On day 2, baseline resting VO_2 was established, then the sprint test series was performed. To control for potential effects on resting VO_2 , this session took place either early in the morning after an overnight fast or 4 hours postprandial with subjects having engaged in minimal activity during the day. Subjects were asked to travel to the laboratory by car, motorcycle or bus

The relationships between aerobic fitness, power maintenance...

Variable	Group		P
	LOW (N=6) Mean (SD)	MOD (N=7) Mean (SD)	
Physical characteristics			
Age (years)	34 (7)	30 (8)	.39
Height (cm)	167.7 (7.6)	167.3 (4.2)	.94
Body mass (kg)	68.6 (7.4)	60.4 (4.8)	.03
VO₂max			
Relative (ml•kg ⁻¹ •min ⁻¹)	34.4 (2.4)	47.6 (3.8)	.00001
Absolute (L•min ⁻¹)	2.36 (.32)	2.86 (.22)	.006
Sprint test series			
Total work: (J•kg ⁻¹)	424 (49)	463 (37)	.14
(kJ)	29.2 (5.7)	27.9 (2.9)	.60
Peak power: (W•kg ⁻¹)	7.8 (1.2)	8.1 (0.8)	.58
(W)	543 (118)	490 (48)	.30
% DO	18.0 (7.6)	8.8 (3.7)	.02

Table 1: Physical and performance characteristics of low (LOW) and moderate (MOD) aerobic power groups.

to eliminate unnecessary activity (Gore & Withers, 1990). Subjects also abstained from caffeine, alcohol, tobacco and drugs for the previous 4 hours and refrained from intense physical exercise for 24 hours prior to this session to minimise the effect on resting oxygen consumption (Short & Sedlock, 1997).

VO₂max test

VO₂max was determined using a Monark friction braked cycle ergometer (model 818) and a continuous maximal oxygen uptake protocol similar to that described by Thoden (1991). Subjects cycled for 2 minutes at 60-80 Watts (W), then power output was increased by 30-40 W every 2 minutes until volitional exhaustion. Using a low-resistance valve (Rudolph 2700) and the breath by breath mode of a Sensormedics Vmax 229 metabolic measurement cart (MMC), expired gases were collected and averaged over 20 s. The gas analysers were calibrated using primary standard reference gases before and after each test. All subjects met the following 3 criteria of VO₂max: a levelling (<2.5 mL•kg⁻¹•min⁻¹ increase) or a decrease in VO₂ with increasing workload, RER in excess of 1.10 and volitional exhaustion. Following 15 minutes of recovery the subject practised 6-second sprints on a cycle ergometer until comfortable with the protocol and confident of producing an all-out effort from a stationary start.

Resting VO₂ and the sprint test series

Upon arrival at the laboratory, the subject rested for 30 minutes with the average VO₂ over the last 10 minutes taken as resting VO₂ (Short & Sedlock, 1997). Resting VO₂ was used to calculate VO₂ above pre-exercise levels. Next, subjects performed a low-intensity warm-up consisting of stretching, then cycling for 5 minutes at 50-60 rpm against a resistance of 0.5 kp, followed by a moderate intensity warm-up consisting of two 10-s sprints at 85 and 115

rpm against a resistance of 1.0 kp, separated by 60 seconds of recovery. Finally, a five-minute stretching period completed the warm-up period. A similar warm up has previously been shown to result in only minor metabolic disturbances (Gaitanos, Williams, Boobis & Brooks, 1993).

The sprint test series was carried out on a friction-braked cycle ergometer (model 818, Monark), loaded at $.075 \text{ kp} \cdot \text{kg}^{-1}$ body weight and interfaced with an electronic revolution counter (Micro Projects) which enabled power output to be monitored and recorded. The product of flywheel revolutions and load were used to determine work and power throughout each exercise period. Total revolutions over each 6-s sprint interval will be used to calculate average power and the best average 6-s power will be referred to as peak power (PP). The percent drop-off from highest average 6-s power occurring in one of the first 3 6-s sprints to the lowest average 6-s power in one of the final 3 6-s sprints will be referred to as percent drop-off (% DO).

The sprint test series consisted of 10 all-out 6-s cycle sprints interspersed with 30 seconds of rest-recovery. To standardise the procedures, each sprint was initiated from a stationary start and subjects remained seated during all sprints, with feet secured to the pedals by toe clips. Subjects were encouraged to produce an all-out effort on each sprint and to avoid breath holding during exercise, as this has been shown to affect the ventilatory response (Fujihara, Hildebrandt & Hildebrandt, 1973), which could impact the distribution of oxygen consumption. The rest periods between sprints consisted of rest-recovery, with the subject seated quietly on the cycle ergometer. During the sprint test series, VO_2 was monitored continuously utilising the breath by breath mode of the MMC then averaged for each 36-s sprint-recovery cycle. Sprint-recovery VO_2 will always be reported as the VO_2 above resting levels (sprint-recovery VO_2 minus resting VO_2).

Statistical procedures

Relationships between variables were assessed using Pearson product-moment correlation coefficients. A two-way ANOVA with repeated measures was used to compare LOW versus MOD groups on performance variables (6-s power and net VO_2 per sprint-recovery period) during the 10 repeats of exercise, then appropriate post hoc analysis was used to follow up significant F values. Student t-tests were used to discern differences between MOD and LOW in physical characteristics, total work (TW), PP, % DO and $\text{VO}_{2\text{max}}$. Statistical significance was accepted at the .05 level. All results are presented as means (standard deviation).

Results

While LOW and MOD differed significantly in $\text{VO}_{2\text{max}}$ when expressed either in absolute or relative terms, there were no significant differences between groups in total work (TW) or peak 6-s power (PP) (Table 1). As seen in Figure 1 while both groups generated similar 6-s power during the first 6 sprints, the MOD group maintained higher power output in each of the final 4 sprints ($p < .05$), resulting in a smaller decrement in % DO for the MOD group over the sprint test series (Table 1).

The MOD group utilised more oxygen during 9 of the 10 sprint-recovery cycles ($p < .05$, Figure 2). No significant relationship was found between % DO and the increase in VO_2 above pre-exercise levels during the sprint-recovery

The relationships between aerobic fitness, power maintenance...

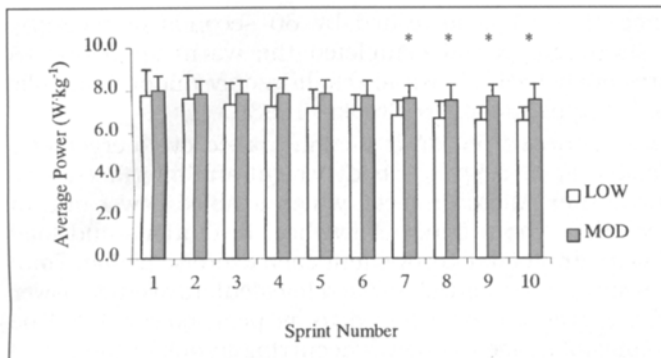


Figure 1: Average power per sprint (mean (SD)) over 10 6-s sprints for moderate (MOD) and low (LOW) aerobic power groups (* $p < .05$).

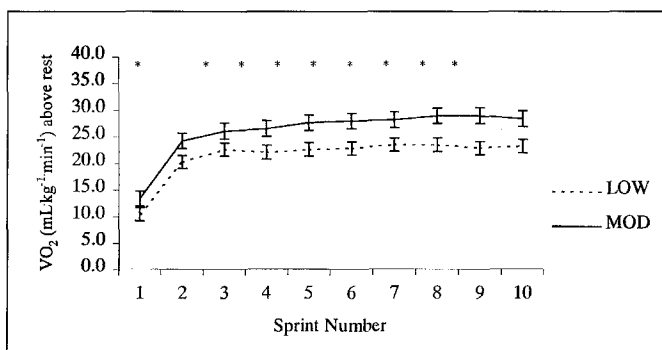


Figure 2: Mean (SD) oxygen consumption above pre-exercise levels per sprint-recovery cycle for moderate (MOD) and low (LOW) aerobic power groups (* $p < .05$).

cycles ($r = -.41$, $p = .15$). However, stronger relationships were found between % DO and VO_2max ($r = -.65$, $p = .02$) and between VO_2max and the aerobic response to the sprint-recovery cycles ($r = .78$, $p = .002$).

Discussion

The primary purpose of this study was to examine the relationships between VO_2max and both power maintenance and oxygen consumption during repeated high intensity exercise within a group of female recreational soccer players. The VO_2max of the LOW group averaged 34.4 (2.4) $\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ while the MOD group averaged 47.6 (3.8) $\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ ($p = .00001$). Whereas the VO_2max of the LOW group is slightly higher than untrained females, the VO_2max values for the MOD group are comparable and/or slightly lower than reported values for elite female soccer players which range from averages of 47.1 to 57.6 $\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ (Colquhoun & Chad, 1986; Jensen & Larsson, 1992; Rhodes & Moshier, 1992).

To minimise the effect of the difference in weight between groups ($p = .03$) results are expressed per unit of body weight. No significant difference in resting VO_2 was seen between the LOW (3.3 (0.5) $\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) and MOD (3.8

(0.7) $\text{mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$) groups so, to make the data more comparable to the literature, net oxygen consumption will be used when discussing the aerobic response to the sprint-recovery cycles, unless otherwise specified. The LOW and MOD groups generated similar peak power. However the MOD group was more successful at maintaining power over the repeated sprints, as seen by their lower % DO (Table 1). Furthermore, as seen in Figure 2, the MOD group consumed significantly more oxygen than LOW throughout 9 of the 10 sprint-recovery bouts. Similar results were shown by Hamilton et al. (1991) when the aerobic response and mean power fatigue of male games players (GP) and endurance trained (ET) athletes over 10 treadmill sprints were evaluated. While assignment to GP and ET groups was on the basis of sport rather than VO_2max , creating a VO_2max overlap between groups, VO_2max was considerably higher in the ET ($60.8 (4.1) \text{ mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$) than the GP group ($52.4 (4.9) \text{ mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$; $p<.01$). As in the present study, both ET and GP groups achieved similar peak power ($p>.05$), yet the ET consumed more oxygen over the 10 sprint-recovery cycles ($p<.05$) and demonstrated a significantly smaller decrement in power over the 10 sprints ($p<.05$). When interpreting their results, Hamilton et al. (1991) suggest that the GP tend to produce higher peak power, despite the fact that the difference between ET and GP was not statistically significant, attributing the high peak power and high % DO to an adaptive response of sprint training. Given the same peak power performance, it would seem more appropriate to suggest that the high % DO in the GP is more likely associated with their lower VO_2max and/or depressed aerobic response to the sprints.

In the present investigation, a moderate relationship of $r = -.65$ ($p = .02$) was seen between VO_2max and % DO over the 10 sprints. Whereas Hamilton et al. (1991) did not compare these variables, Dawson et al. (1993) and McMahon and Wenger (1998) found similar relationships between VO_2max and power decrement during six 6-s cycle sprints ($r = -.56$, $p<.05$) and six 15-s cycle sprints ($r = -.62$, $p=.002$), respectively. Hamilton et al. (1991) reported a moderately high correlation between VO_2max and the aerobic response to repeated sprint recovery intervals ($r = .83$; $p<.01$), which is confirmed in the present investigation ($r = .78$; $p=.002$) and suggests that the aerobic response to repeated bouts of brief high intensity effort is related to VO_2max . Similarly, in subjects who completed two 30 s all-out cycle sprints, Bogdanis et al., al. (1996a) found a high correlation between VO_2max and the percent of energy contribution by aerobic metabolism on sprint 1 ($r=.79$) and sprint 2 ($r=.87$). It would seem that the higher the maximal oxygen uptake the greater the likelihood of the athlete using more oxygen during the sprint-recovery intervals. In support of this, the average gross sprint-recovery VO_2 was higher in MOD than LOW (29.8 versus $24.6 \text{ mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$, respectively; $p=.02$) even though this represented a higher percentage of VO_2max in LOW subjects (LOW vs MOD: 71.6 (4.8) versus 62.6 (5.9) % VO_2max ; $p=.01$). VO_2max thus appears to determine the magnitude of the aerobic response to repeated sprints, which in turn may dictate the extent of recovery.

Even a single 6-s bout of high intensity exercise is enough to lower ATP/PCr stores significantly, elevate blood and muscle lactates and depress pH (Gaitanos et al., 1993). With subsequent exercise bouts, as pH decreases further through increases in anaerobic lactate metabolism, contractile

processes are disrupted. Furthermore, restoration of PCr levels during recovery is critical for maintenance of power during repeated 6-s sprints (Gaitanos et al., 1993) as it provides the maximum rate of ATP resynthesis and helps to minimise lactate production. Bogdanis, Nevill, Boobis, Lakomy & Nevill (1995) found that power recovery on repeated 30-s cycle sprints and resynthesis of PCr proceeded in parallel, confirming the relevance of PCr availability for subsequent performance. Complete repletion of ATP/PCr stores may require 3-5 minutes (Hultman, Bergstrom & McLennan-Anderson, 1967) or even longer (Harris, Edwards, Hultman, Nordesjo, Nylinde & Sahlin, 1976), so, with only 30 seconds of recovery between exercise bouts, ATP/PCr stores will only be partially restored. Furthermore, with subsequent high intensity work bouts, as in the present study, ATP/PCr will be progressively depleted (Gaitanos et al., 1993; Yoshida & Watari, 1993) and there will be increased reliance on anaerobic glycolysis (Wootton & Williams, 1983), which may adversely affect performance (Sahlin, 1992). Neither blood or muscle lactates were measured in the present investigation. However, using a similar protocol, Hamilton et al. (1991) found that following multiple treadmill sprints the ET, who had similar peak power to the GP ($p > .05$), had a smaller performance decrement ($p < .05$), consumed more oxygen ($p < .05$) and had lower blood lactate levels ($p < .05$) than the GP, suggesting smaller contribution from anaerobic glycolytic energy production, enhanced lactate removal or a combination of both in the ET. Endurance training results in lower blood and muscle lactates for the same absolute submaximal workload (Karlsson, 1971) due to decreased production of lactate as a result of increased reliance on other energy systems (Holloszy & Coyle, 1984) and/or increased lactate clearance (Brooks & Donovan, 1983).

PCr recovery appears to be dependent on oxygen supply to the muscle (Harris et al., 1976) and has been coupled to oxygen consumption in the immediate post-exercise period (Hultman et al., 1967; Piiper & Spiller, 1970). Therefore the superior levels of VO_2 seen in the MOD group throughout the sprint-recovery cycles may contribute to PCr recovery, which may partially explain why the MOD group were more successful at maintaining power throughout the 10 sprint repeats. Balsom, Gaitanos, Ekblom and Sjodin (1994) found that, in subjects performing 10 6-s sprints under normoxia and hypoxia conditions, during normoxia conditions subjects consumed more oxygen, accumulated less blood lactate and demonstrated superior power maintenance, which implies either increased lactate clearance or, more likely, an increased reliance on phosphagen and/or aerobic energy sources with improved oxygen delivery. Additionally, Balsom, Ekblom & Sjodin (1994) found that by inducing increases in haemoglobin, which also increased $\text{VO}_{2\text{max}}$, subjects performing fifteen 6-s treadmill sprints showed reduced accumulations of blood lactate and the adenosine degradation product, hypoxanthine, despite performing the same amount of exercise when compared to the control condition.

While significant relationships were found between %DO and $\text{VO}_{2\text{max}}$ ($r = -.65$, $p = .02$) and between $\text{VO}_{2\text{max}}$ and net sprint-recovery VO_2 ($r = .78$, $p = .002$), the correlation between %DO and net sprint-recovery VO_2 ($r = -.41$, $p = .16$) was not significant. Hamilton et al. (1991) reported a modest correlation ($r = -.60$, $p < .05$) between %DO and net sprint-recovery VO_2 . Perhaps factors

affecting VO_2max impact somewhat differently on the %DO than the aerobic response to exercise. It may be that rate of recovery is determined more by genetic factors such as the percentage of slow twitch fibres, whereas the aerobic response to exercise is influenced by factors which are sensitive to training. In support of this, Colliander, Dudley and Tesch (1988) demonstrated that individuals in a "low fast twitch" group were superior to a "high fast twitch" group in restoring force between sets of concentric contractions. Furthermore, the oxidative capabilities of the muscle can be enhanced (Jansson & Kaijser, 1977), but the enhancements do not always translate into improvements in VO_2max (Holloszy & Coyle, 1984). It may also be that the aerobic response is more tightly linked to aerobic capacity than VO_2max . Bogdanis et al. (1996a) found a relationship between aerobic capacity, as characterised by 4 mmol•L blood lactate (Tanaka & Matsuura, 1984) and power output recovery ($r=.75$) and between PCr resynthesis and power output recovery ($r=.94$). Unfortunately no relationship was reported between VO_2max and power recovery or power recovery and PCr restoration.

In summary, a higher maximal aerobic power is associated with enhanced aerobic contribution and superior power maintenance during repeated supramaximal cycle sprints in female recreational soccer players. Since the aerobic fitness of the subjects in the present investigation is moderate, caution should be used in extrapolating these results to the highly trained. While it is likely that a more highly endurance trained group would have superior power maintenance and utilise more oxygen on repeated sprints, as implied by the work of Hamilton et al. (1991), the possibility of an aerobic fitness threshold exists, beyond which improvements in VO_2max do not translate into further enhancements of recovery. These findings support the practice of enhancing low to moderate levels of VO_2max so as to improve performance on repeated bouts of high intensity activity.

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